Traduce Señas:

User Manual

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1. Purpose of the System

The images in this manual are real screenshots of the application with fictitious data. They show the app in Spanish; the labels quoted in this manual are the ones the app shows when its language is set to English.

Traduce Señas is an application (web and, partly, mobile) that uses the camera to recognize **Colombian Sign Language (LSC)** and show what it understands as **text**. That text can also be **heard as speech**.

2. Scope of the System

What it does today:

- Recognizes signs with the camera and turns them into **text**.
- Reads the text **aloud**.
- Saves what was translated in a **history** that you can view and delete.
- Offers a **daily challenge** for translating and gives you **points** and **medals**.

What it does **not** do (not in this version):

- It does not turn text or speech into signs. There is no 3D avatar that performs the signs either.
- It does not replace a **certified interpreter**. It can make mistakes. For important matters (health, legal procedures, formal education), use an interpreter.
- It only knows a limited vocabulary: about 54 words and the alphabet (see section 6).

3. Requirements

Table 1

Requirements

You need	Details
A device with a camera	A computer with a webcam, or a phone.

You need	Details
A recent browser or the phone app	Camera translation works in the web version. On a phone, **Words** mode opens inside the app and needs internet.
Internet	The app downloads the recognition tools (MediaPipe; Zhang et al., 2020) from the internet when the camera opens.
Camera permission	The browser or the phone asks for it the first time. You must accept it.
Good lighting	Light from the front or from above; avoid windows or lamps behind you.
Good framing	Your whole hands must be visible, with space on both sides.
An email address	To create the account and confirm it with a code.

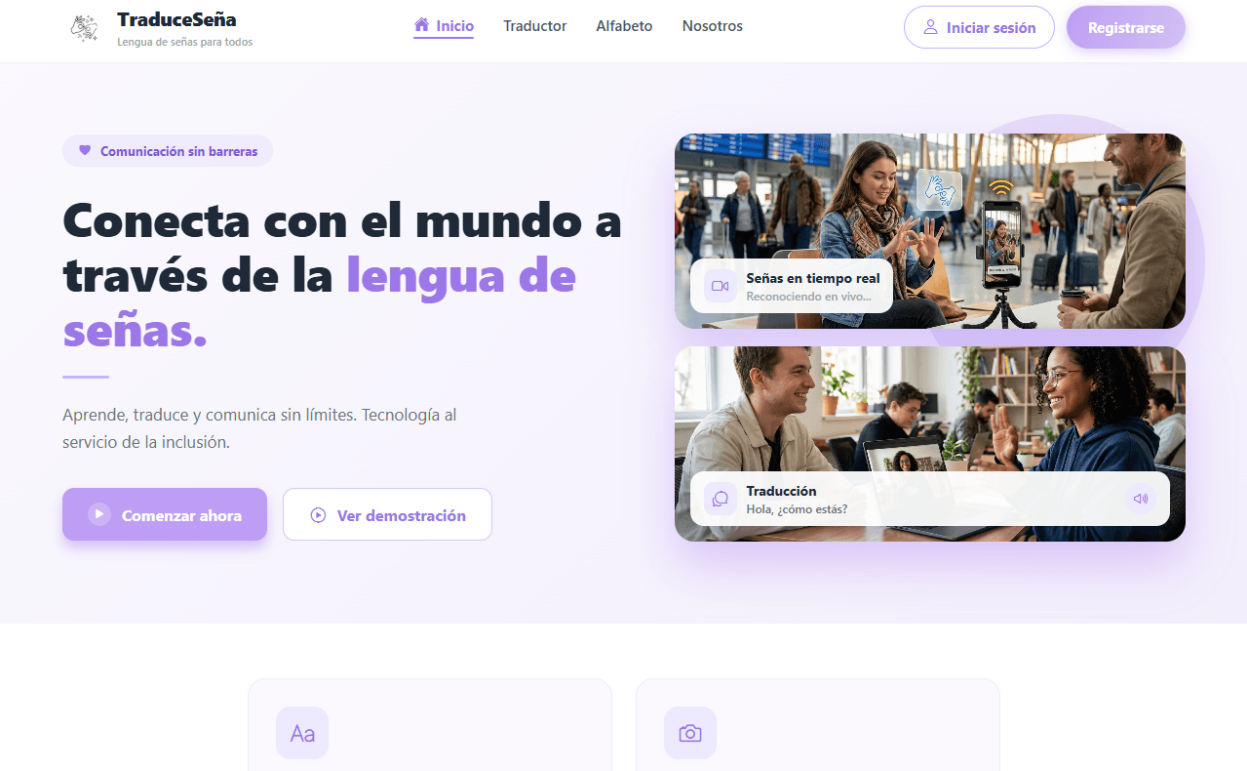
4. Accessing the System

4.1 Opening the Application

1. Open the address given to you by whoever manages the application. In a normal local installation it is `http://localhost:8081` with the development web app, or the public address of the application (see the Installation manual).
2. You will see the home page with the **Sign up** and **Sign in** options. On the web there are also presentation pages (About us, translator demo and alphabet demo).

Figure 1

Application Home Page

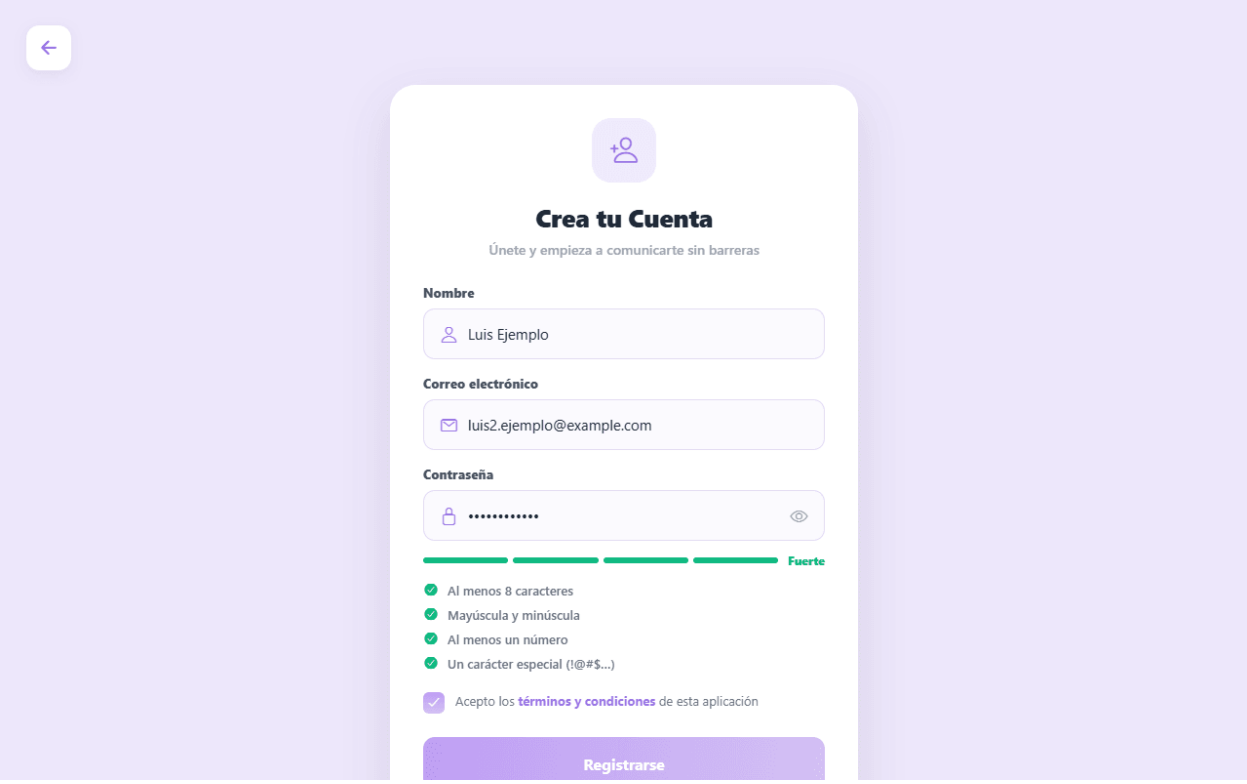


4.2 Signing Up

1. On the home page, tap **Sign up**.
2. On **Create your Account**, enter your full name, your email address and a password.
3. The password must have **at least 8 characters** and meet at least **three** of these four types: uppercase letters, lowercase letters, numbers and symbols. The strength indicator tells you whether it is Weak, Medium or Strong. To be Strong it needs 12 characters or more and all four types.
4. Check **I accept the terms and conditions**. You cannot continue without it.
5. Tap **Sign up**.

Figure 2

Create Your Account Screen, With Fictitious Data



If the email already has an account you will see "This email already has an account".

4.3 Confirming Your Email

Your account starts out **inactive** until you confirm your email.

1. Check your email (and the spam folder). You will receive a **6-digit code**.
2. On the **Confirm your email** screen, enter the 6 digits and tap **Confirm**.
3. When "Email confirmed, please sign in" appears, you can sign in.

Figure 3

Confirm Your Email Screen



Things you should know:

- The code **expires after 15 minutes**.
- You have a maximum of **5 attempts** per code. If you get it wrong many times you have to request another one.
- To request another code, use **Resend**. You must wait **60 seconds** between one send and the next, and **5 sends per hour** are allowed.
- If you close the page before confirming, when you come back you will see "Sign in to finish confirming your email".
- If "We couldn't send the code, request a new one" appears when you sign up, the account was created: request the code with **Resend**.

4.4 Signing In and Out

1. Tap **Sign in**.
2. Enter your email and password and tap **Sign in**.

Figure 4

Sign In Screen



Possible messages:

Table 2

Signing In and Out

Message	What it means
"Incorrect email or password"	Check what you typed.
"Confirm your email to continue"	The code from step 4.3 is missing.
"Your account is blocked. Contact support."	An administrator blocked the account.
"Could not reach the server..."	Check your internet connection or tell the administrator.

Message	What it means
"Your session has expired. Please sign in again."	The session has a fixed length and does not renew by itself: sign in again.

Google and Facebook. These buttons work when the application has that sign-in configured. If a button appears disabled or says it is not configured, sign in with email and password. With Google, if the account is new, the app asks you to accept the terms before creating it.

To **sign out**: go to the **button with your name**, at the top right, and tap **Sign out**.

5. Functions of the User Role

5.1 Translating Signs to Text

After signing in you will see the bottom bar (or top bar on wide screens) with the **Translation**, **Alphabet** and **History** tabs, and the **button with your name** (**Profile**) at the top right.

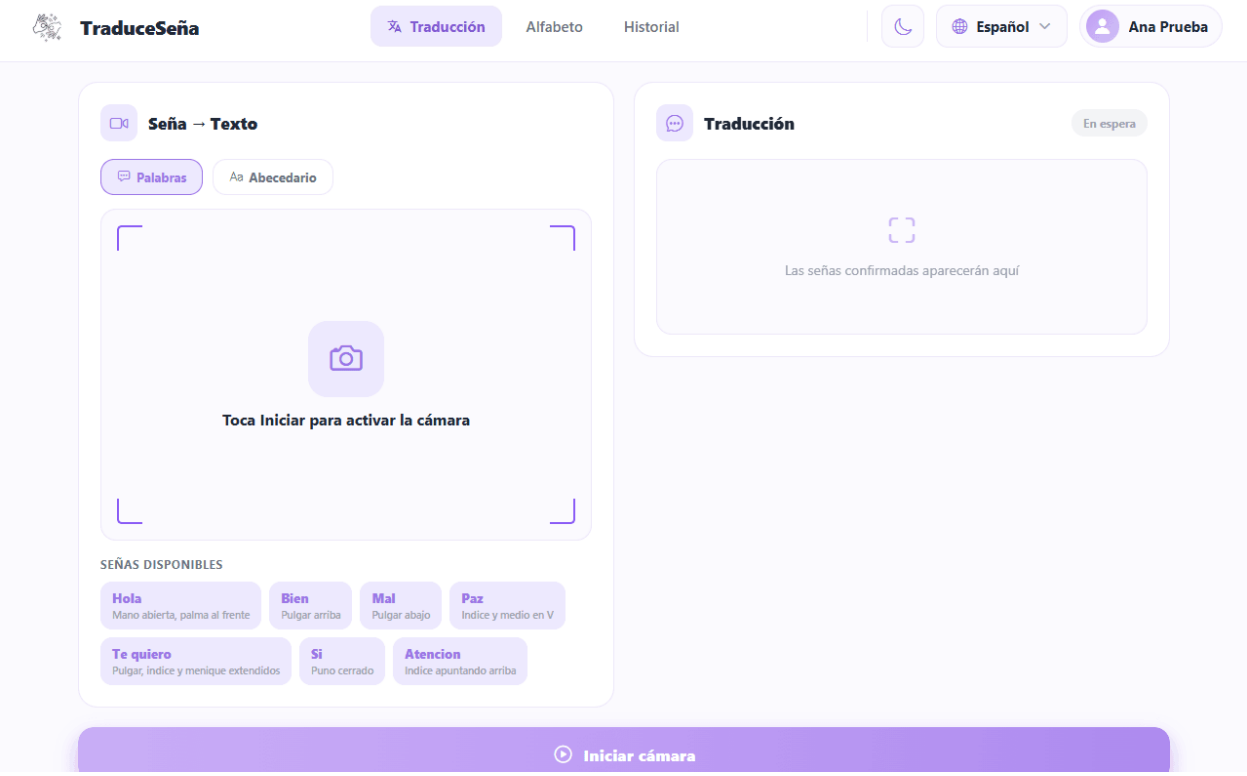
If your account is an administration account you will see other tabs (Statistics and Admin) and not the translation ones. This is explained in the Administrator manual.

The **Translation** screen has two modes, which you choose with two buttons:

- **Words**: recognizes complete LSC words. It is the default mode. In the web version it uses the computer camera; on a phone it opens inside the app and needs internet.
- **Alphabet**: recognizes letters (fingerspelling). **It only works in the web version.** On a phone you will see the notice "Sign recognition is not available on mobile yet. Use the web version to translate with the camera".

Figure 5

Translation Screen, Words Mode



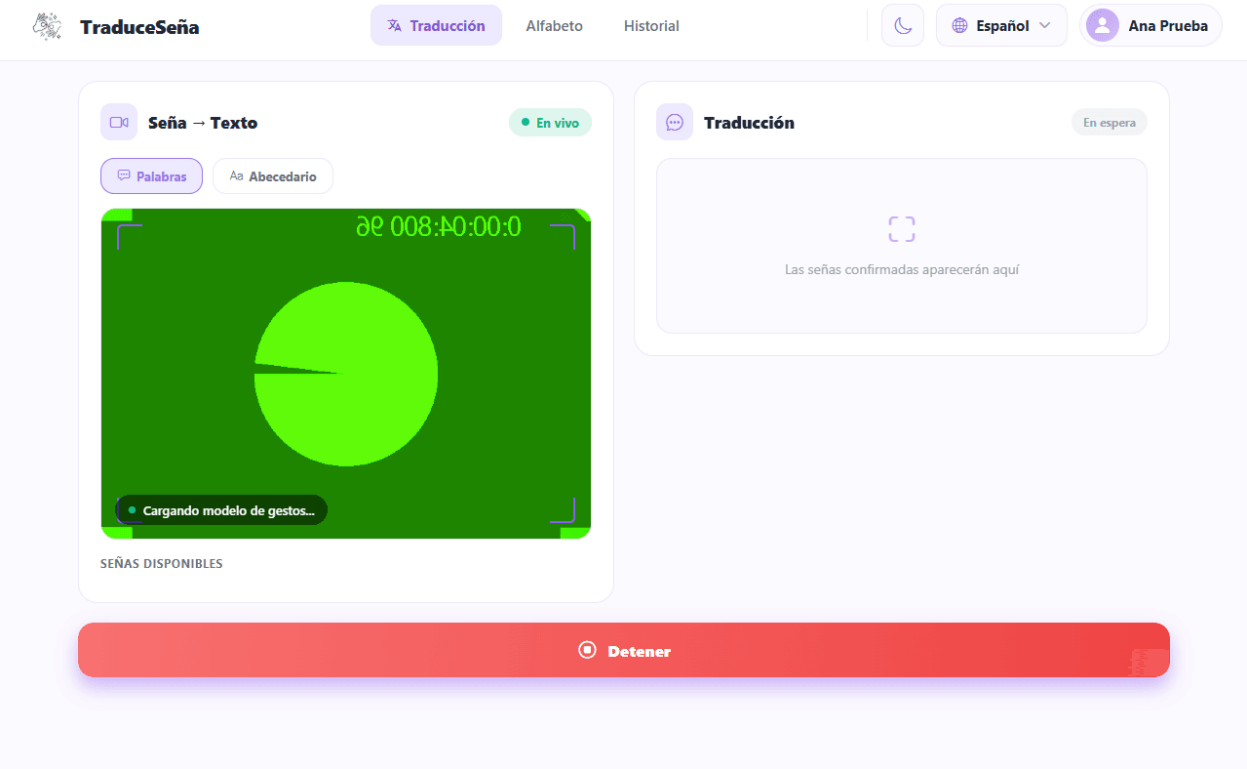
5.1.1 Words Mode

1. Choose **Words**.
2. Tap **Start camera**. If the browser or the phone asks for permission, accept it.
3. Wait for "Loading gesture model..." to finish (the first time takes longer).
4. Stand in front of the camera with your hands in view. You will see "Show a sign to the camera".
5. Make the **complete sign, clearly and calmly**. The screen shows "Signing…", "Analysing…".
6. When the app is confident, the word appears in **Confirmed signs**. Each confirmed sign is **saved to the history automatically**.

7. Tap **Stop** when you are done.

Figure 6

Camera Active; the Camera Image Is Simulated



Tips from the app itself: keep your hand centered in the camera, ensure good lighting and repeat the sign if it is not detected.

If you see "Low confidence, try the sign again" or "Could not recognise", repeat the sign more calmly. If you see "No hands in frame", move closer or center your hands.

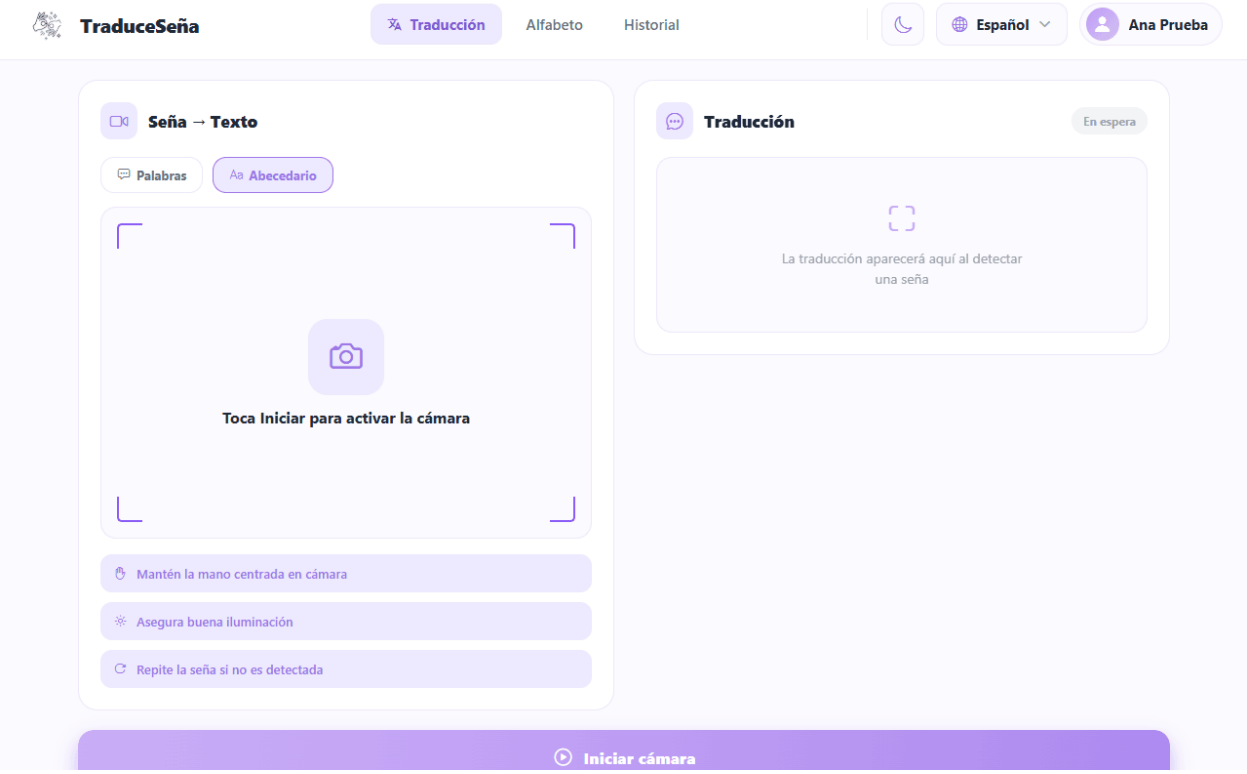
5.1.2 Alphabet Mode

1. Choose **Alphabet** and tap **Start camera**. This mode works in the web version.
2. Make the letter in front of the camera and hold it for a moment.
3. The recognized letters are added to **Recognized text**.

4. Use **Space** to separate words, **Delete** to remove the last letter and **Clear** to start over (the trash can deletes the confirmed signs of Words mode).
5. Tap **Stop** when you are done.

Figure 7

Alphabet Mode Selected



When the app shows "Confidence", it is how sure it is about what it read: higher is better.

5.2 Listening to and Copying the Result

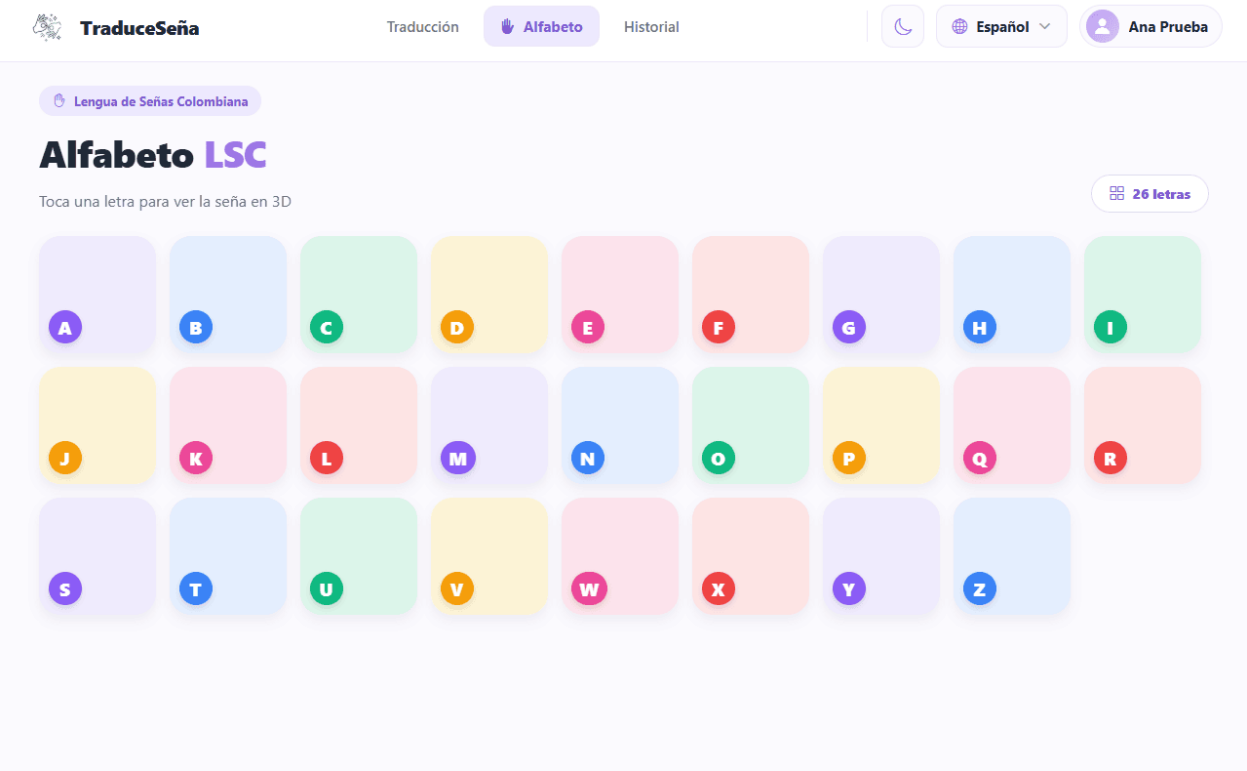
- Tap the **speaker icon** (below the result) to have the device read the text aloud. The voice uses the language you chose in Profile.
- Tap the **copy icon** to copy the text to the clipboard. You will see "Text copied to clipboard".

5.3 Alphabet

The **Alphabet** tab shows the letters. Tap a letter to see its description and the sign. It is useful for practicing before using Alphabet mode.

Figure 8

LSC Alphabet

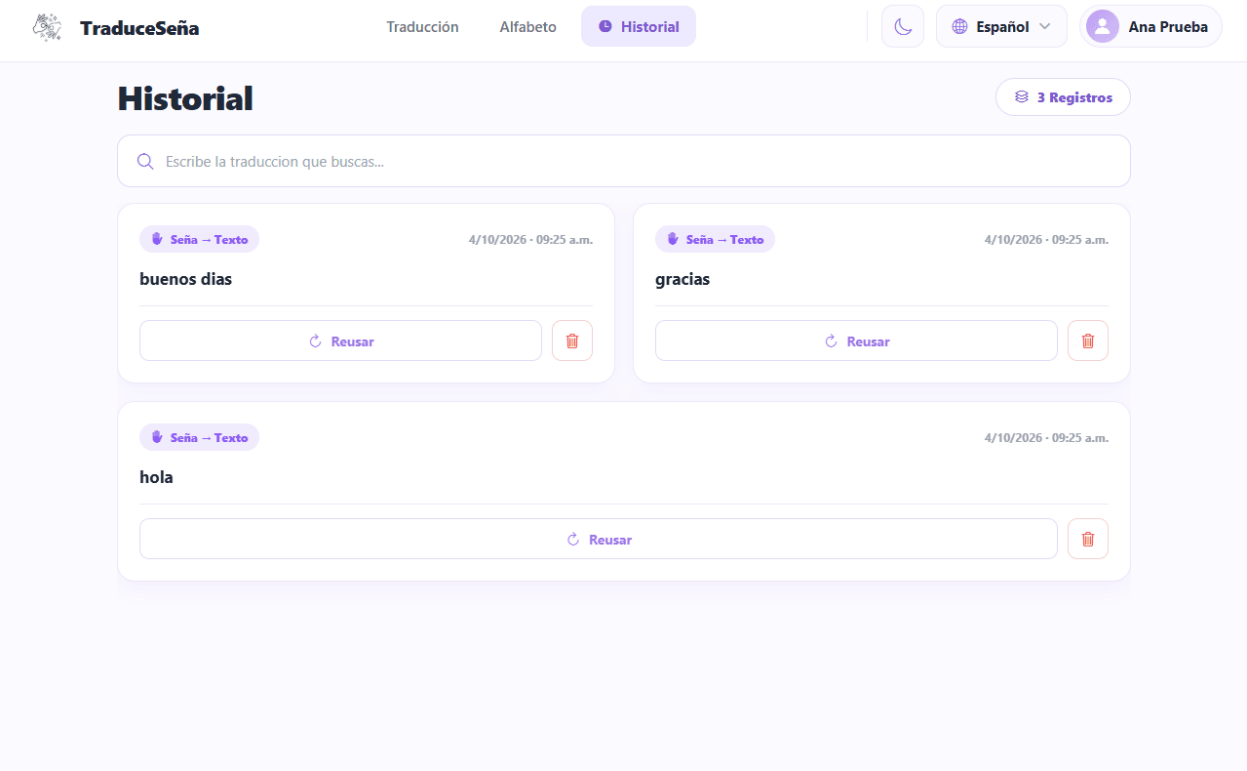


5.4 History

The **History** tab lists your saved translations (up to the last 100).

Figure 9

History With Example Translations



1. Type in the search box ("Type in the translation you are looking for...") to filter.
2. To **delete** one, tap the **trash can icon** on that record and confirm "Do you want to delete this history record?".
3. If you have not made any translations yet you will see "No translations".

Tapping **Reuse** on a record shows "Reuse translation": for now it only shows a notice with the text; it does not translate again.

Deleting removes the record from your history (the app marks it as deleted).

The history is shown grouped by sessions: each card gathers the translations you made in one session. Also, translations are deleted automatically 30 days after they are saved, so the history only keeps the last month.

5.5 Profile

In **Profile** you can:

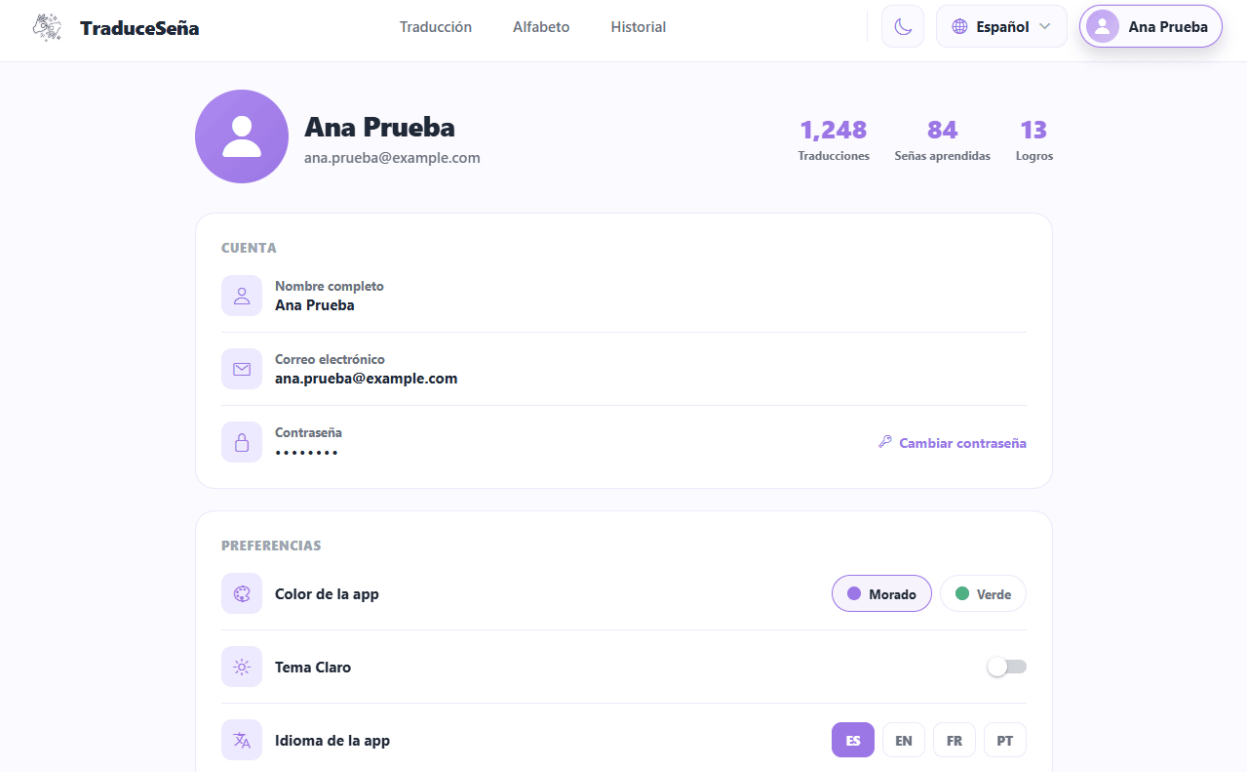
Table 3

Profile

Option	What it does
Theme (Light / Dark)	Changes the app colors.
App language	Spanish, English, French or Portuguese.
Change password	Changes your password. It asks for the current one. If your account was created with Google and has no password, the app sends a code to your email so you can create one.
Terms and conditions / Privacy policy	The app's legal texts. See the notice in section 7.
Achievements	Medals showing your progress, including those earned with challenge points (see 5.6).
Sign out	Leaves your account.
Delete account	Deletes your account and your data. See the following notice.

Figure 10

Profile of a Fictitious Account; the Figures at the Top Are Example Values



Delete account permanently deletes your account. The app asks for your password. If the account was created with Google and has no password, it sends a code to your email and asks for that code. Your account, your history and your progress are deleted. If you are the only administrator, the app does not let you delete the account.

5.6 Daily Challenges and Medals

Every day the app gives you a translation challenge: save 1, 3, 5 or 8 translations. All users see the same challenge of the day, and it changes at midnight (Colombian time).

- In the **Profile** you will see a card with your progress for the day. It updates every time you save a translation; only saved translations count.
- When you complete the challenge you earn points (5, 10, 15 or 25, depending on the target) and a notice appears on screen. Points are counted only once.
- The Profile shows the total of your challenge points.
- With the accumulated challenge points you unlock **medals** in **Achievements**. When you unlock one, a notice appears.
- If you close a notice, your points and your completed challenge are saved just the same.

6. Known Limitations

- **Short vocabulary.** The word model knows 54 words, plus the alphabet. It does not recognize anything else.
- **Camera and light.** The model learned from people recorded in a studio. With a different camera, a different distance or poor light it may fail even if you make the sign correctly.
- **It can be wrong.** Always check the text. Do not use the app as the only source in important situations.

- **Alphabet only on the web.** On a phone, Alphabet mode does not recognize signs with the camera. Words mode on a phone needs internet.
- **One direction only.** Only from sign to text or speech.
- **Statistics:** only administration staff can see them.

7. Your Data and Your Privacy

- The app asks for the camera only to recognize signs. Recognition is processed on your device, inside the browser or the app.
- Your account data (name, email, protected password), your translation history and your challenge and achievement progress are stored. Translations are deleted automatically 30 days after they are saved.
- Colombian law (Law 1581 of 2012 and Decree 1377 of 2013) gives you the right to know, correct and request the deletion of your data. You can do it yourself from **Profile**, with **Delete account**. Deletion is permanent.
- The privacy texts shown by the app have not been checked against how this version actually works; this manual does not repeat their claims.

8. If Something Goes Wrong

Table 4

If Something Goes Wrong

Problem	What to do
The camera does not turn on	Accept the camera permission in the browser (padlock icon next to the address) or on the phone. Close other applications that use the camera.
"Loading model..." never finishes	Check your internet connection and reload the page.
It does not recognize the sign	More light, hands centered, repeat calmly. Move a little further away or closer.
The code does not arrive by email	Check spam; wait 60 s and use **Resend**. Maximum 5 per hour.

Problem	What to do
"Incorrect or expired code"	Request a new code and use it within 15 minutes.
"Your session has expired"	Sign in again.
The Google or Facebook button does nothing	That sign-in is not configured in the app. Sign in with email and password.
The app does not open	Tell the administrator: the service may be down.

9. Minimal Glossary

Table 5

Minimal Glossary

Word	Simple meaning
LSC	Colombian Sign Language (Lengua de Señas Colombiana).
Sign	A hand gesture that means a word or a letter.
Alphabet / fingerspelling	Spelling with the hands, letter by letter.
Confidence	How sure the app is about what it recognized.
History	The list of your saved translations.
Session	The time you are signed in to your account. It has a fixed length and does not renew by itself.
Certified interpreter	An accredited LSC professional. This app does not replace one.

References

Decreto 1377 de 2013 [Decree 1377 of 2013]. *Por el cual se reglamenta parcialmente la Ley 1581 de 2012* [Partially regulating Law 1581 of 2012]. June 27, 2013.

Ley 1581 de 2012 [Law 1581 of 2012]. *Por la cual se dictan disposiciones generales para la protección de datos personales* [Issuing general provisions for the protection of personal data]. October 17, 2012. D.O. No. 48587.

Zhang, F., Bazarevsky, V., Vakunov, A., Tkachenka, A., Sung, G., Chang, C.-L., & Grundmann, M. (2020). *MediaPipe Hands: On-device real-time hand tracking* [Preprint]. arXiv. https://doi.org/10.48550/arXiv.2006.10214